

ClassNotFoundException in Java

Meaning of Class Not Found Exception

ClassNotFoundException is a checked exception in Java. It occurs when a program tries to load a class at runtime, but the Java Virtual Machine (JVM) cannot find the required class. Here, **Class.forName("UnknownClass")** tries to load a class that is not available.

Objective

To demonstrate how Java handles ClassNotFoundException when a class cannot be found during runtime, using Class.forName() and a try-catch block.

Task Description

Write a Java program that attempts to load a non-existing class using Class.forName(). Catch the resulting ClassNotFoundException and display an appropriate message instead of allowing the program to terminate with an unhandled exception.

Step-wise Execution

1. The program starts execution from the main() method.
2. A try block is used for the statement that may cause an exception.
3. Class.forName("UnknownClass") attempts to load the class at runtime.
4. The class cannot be found, so Java throws ClassNotFoundException.
5. Control moves to the catch block.
6. The catch block prints "Exception: Class not found".
7. The program ends normally after handling the exception.

Algorithm

1. Start.
2. Enter main().
3. Begin try block.
4. Load UnknownClass using Class.forName().
5. If the class is unavailable, catch ClassNotFoundException.
6. Display the exception message.
7. Stop.

UML Diagram



Flow: Start → Class.forName("UnknownClass") → ClassNotFoundException → catch block → Print message → End

Program Code

```
class ClassNotFoundDemo {  
    public static void main(String[] args) {  
        try {  
            Class.forName("UnknownClass");  
        } catch (ClassNotFoundException e) {  
            System.out.println("Exception: Class not found");  
        }  
    }  
}
```

Code Explanation

class ClassNotFoundDemo — Defines the program class.

main() — Starting point of execution.

try — Contains code that may throw an exception.

Class.forName("UnknownClass") — Dynamically attempts to load a class by name.

catch (ClassNotFoundException e) — Handles the exception when the class is unavailable.

System.out.println() — Displays the error message.

Output Screenshot

```
C:\> java ClassNotFoundDemo
```

```
Exception: Class not found
```

Expected Output:

```
Exception: Class not found
```